

Pathshield – User’s Guide

The pathshield app allows you to plot safer and more informed navigation routes, influenced by safety and traffic data.

Dependencies

To run this app currently, you will need some dependencies in python. When officially released this will be running on a server so no need for dependencies.

Current python packages to be installed by pip are listed below:

- Joblib
- Osmnx
- Network
- Folium
- Shapely
- Geopandas
- Pandas
- Flask
- Numpy
- Dotenv
- sklearn

once these are installed in the project’s environment, we can continue to run the application.

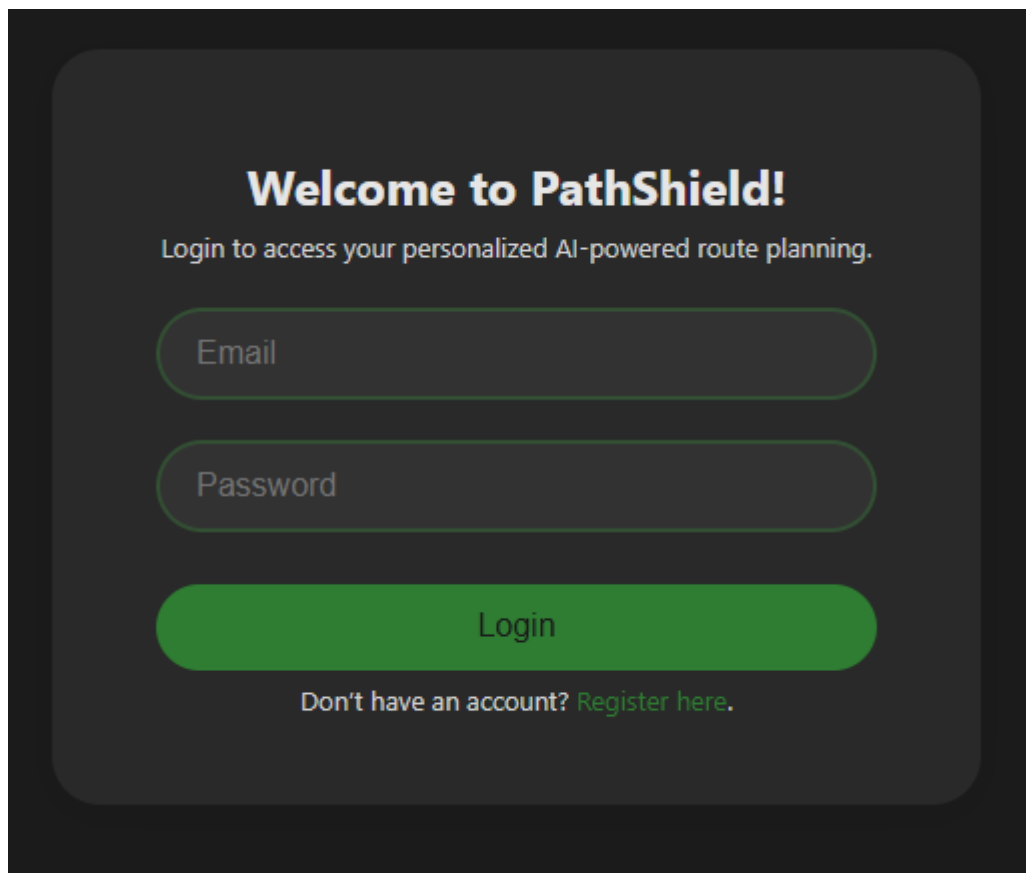
Running the Application

The application hosts on a local host server. To start the server, run “run.py” in the prototype 5 folder, then put

<http://127.0.0.1:5000> in your chrome or preferred browsers search bar while the program is running.

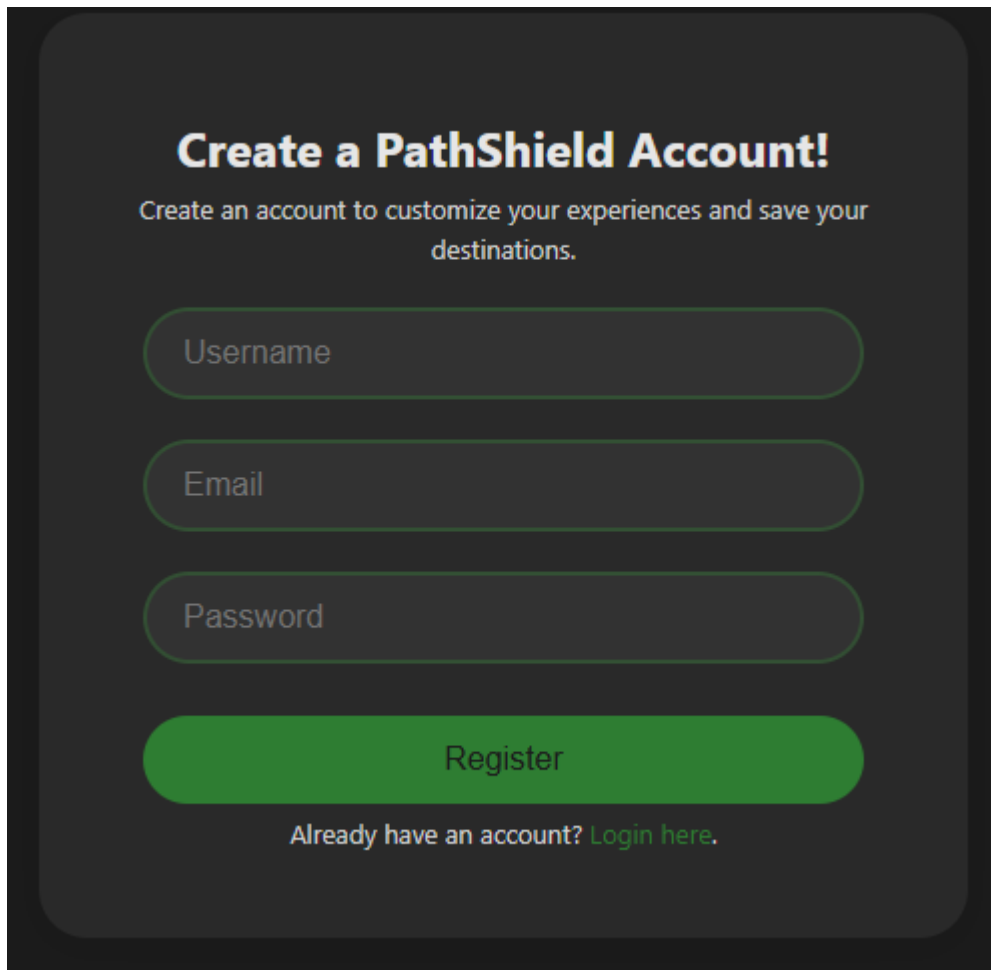


This will take you to this screen:



Navigating the program

If you don't have an account, you can register with the register here link, it will prompt you to enter a username, email, and password.

A registration form titled "Create a PathShield Account!" with a subtitle "Create an account to customize your experiences and save your destinations." It features three input fields for "Username", "Email", and "Password", followed by a green "Register" button. At the bottom, it says "Already have an account? [Login here.](#)"

Create a PathShield Account!

Create an account to customize your experiences and save your destinations.

Username

Email

Password

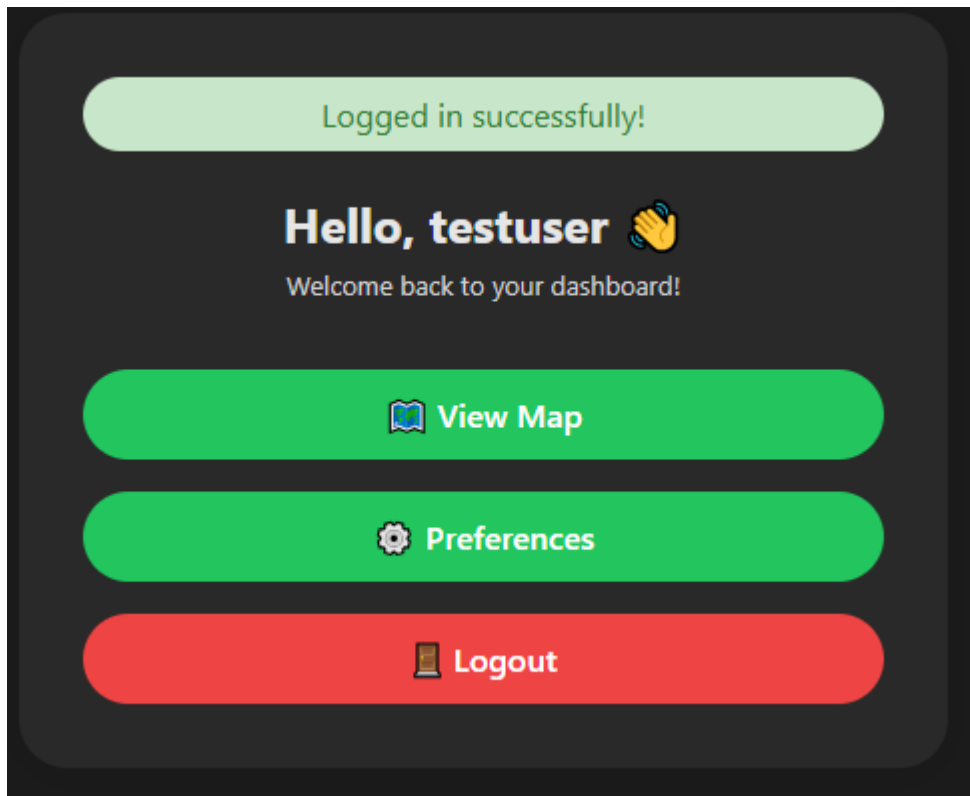
Register

Already have an account? [Login here.](#)

Once you have an account, you can log in using the login screen. Enter your email and password to continue.

(for debugging there is a test user, email: testuser@gmail.com, password: 123456)

Once logged in there is a dashboard with 3 options: logout, preferences, and view map. Logout will exit the program back to the login screen, preferences will take you to settings (these are currently placeholders as for the MVP they were minor features like changing the UI appearance) and view map will take you to the map screen.

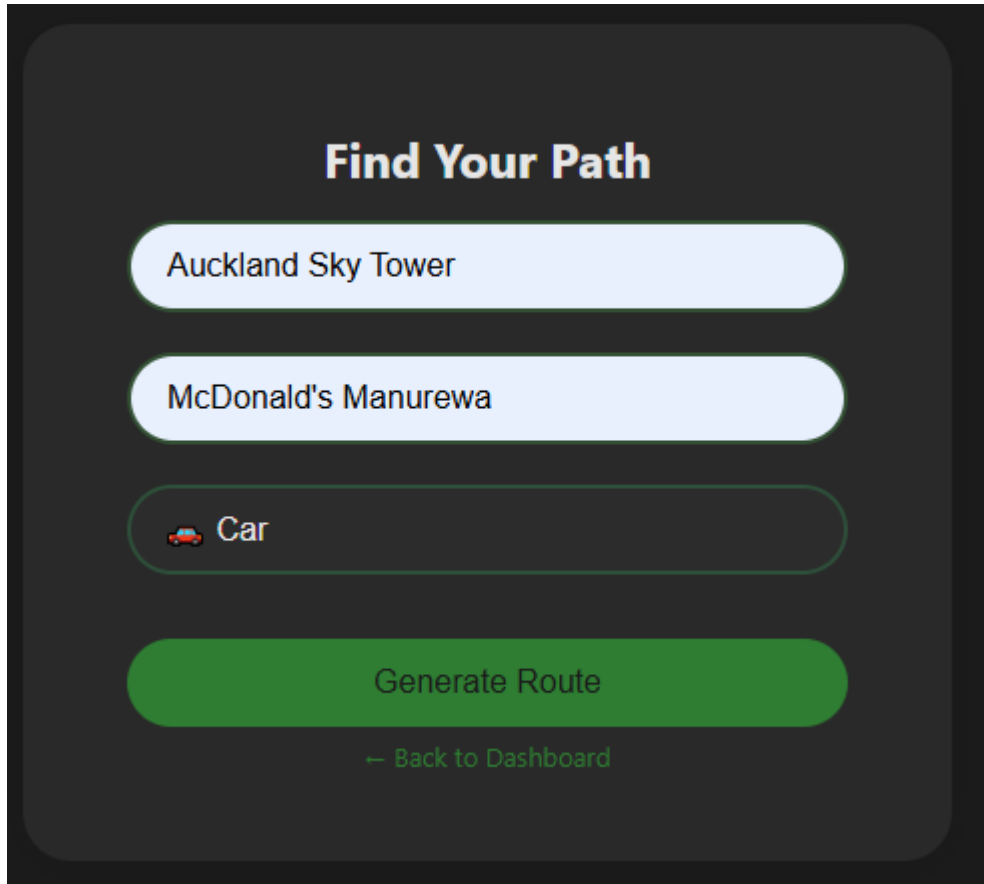


Map Input Screen

The map screen lets you input a starting point and a destination, with a mode of transport (currently for the MVP only driving is supported) the inputs should be names of landmarks or addresses with suburbs (e.g. Sky tower, 45 Colchester Avenue Glendowie, McDonalds Greenlane)

Once you've input your addresses, then click the Generate Route button. It'll load for a few seconds, then move you to the interactive map.

If the address is invalid, then the app will pop up an error and let you reenter the addresses. (addresses must be in the Auckland region)

A screenshot of a mobile application interface titled "Find Your Path". The interface has a dark background with rounded rectangular input fields. The first field contains "Auckland Sky Tower", the second contains "McDonald's Manurewa", and the third contains a car icon followed by the text "Car". Below these fields is a large green button labeled "Generate Route". At the bottom, there is a smaller green button labeled "← Back to Dashboard".

Find Your Path

Auckland Sky Tower

McDonald's Manurewa

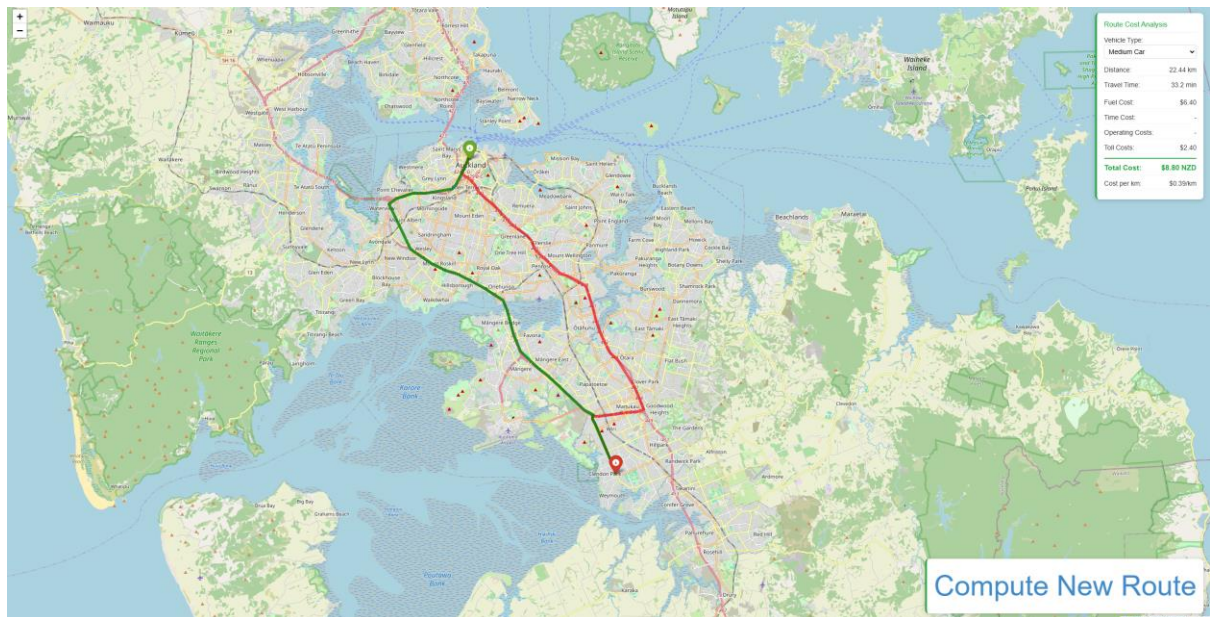
 Car

Generate Route

← Back to Dashboard

Map Screen

The route will be generated and displayed on the map. The red line is the non-machine learning adjusted route, and the green is the route adjusted for high crime areas and traffic. The cost calculator will calculate the cost for the route, as well as the time to get to the destination. You can compute a new route with the “Compute New Route” button.



That's all you need to successfully navigate the app and generate routes!